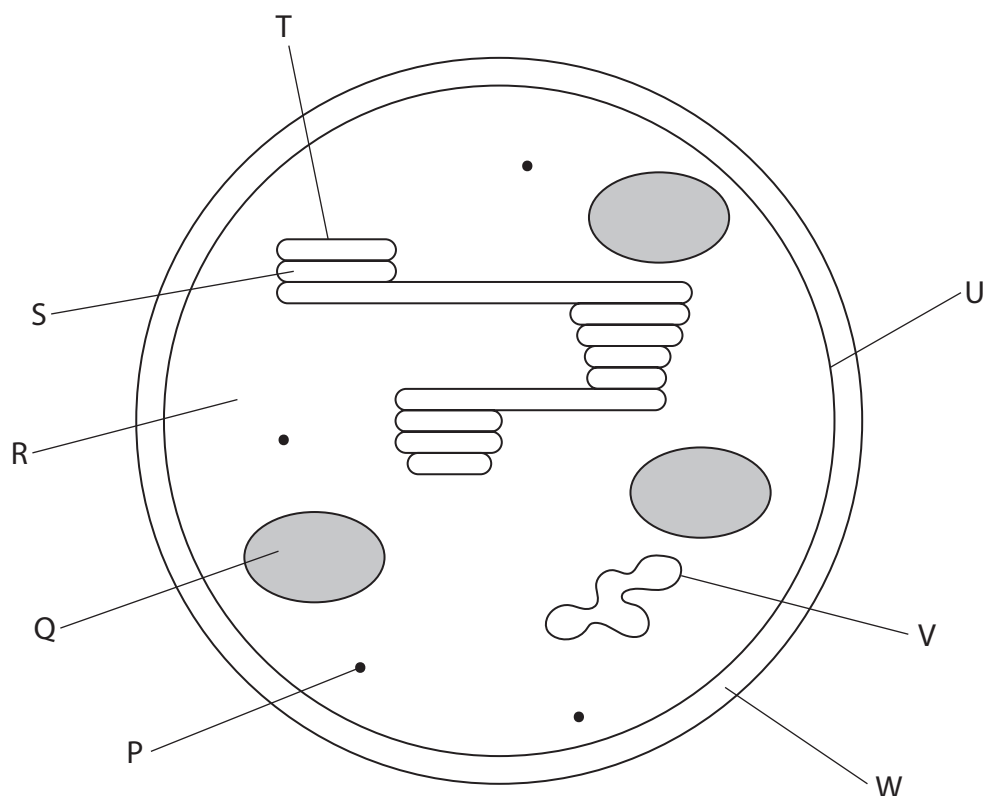


4 The structure of a chloroplast is related to its role in photosynthesis.

(a) The diagram shows a chloroplast.



(i) Which row of the table shows where the light-dependent reactions and the light-independent reactions take place?

(1)

		Light-dependent reactions	Light-independent reactions
☐	A	R	S
☐	B	S	T
☐	C	T	R
☐	D	T	S

(ii) Where do hydrogen ions (protons) accumulate?

(1)

- ☐ **A** Q
- ☐ **B** R
- ☐ **C** S
- ☐ **D** W

(iii) The diagram is 16 000 times larger than an actual chloroplast.

Calculate the surface area of the actual chloroplast.

Use the formula:

$$\text{Surface area} = 4 \pi r^2$$

Give your answer as a whole number in μm^2 .

(3)

Answer μm^2

(iv) Explain the advantages of thylakoids in a chloroplast having a large surface area.

(2)

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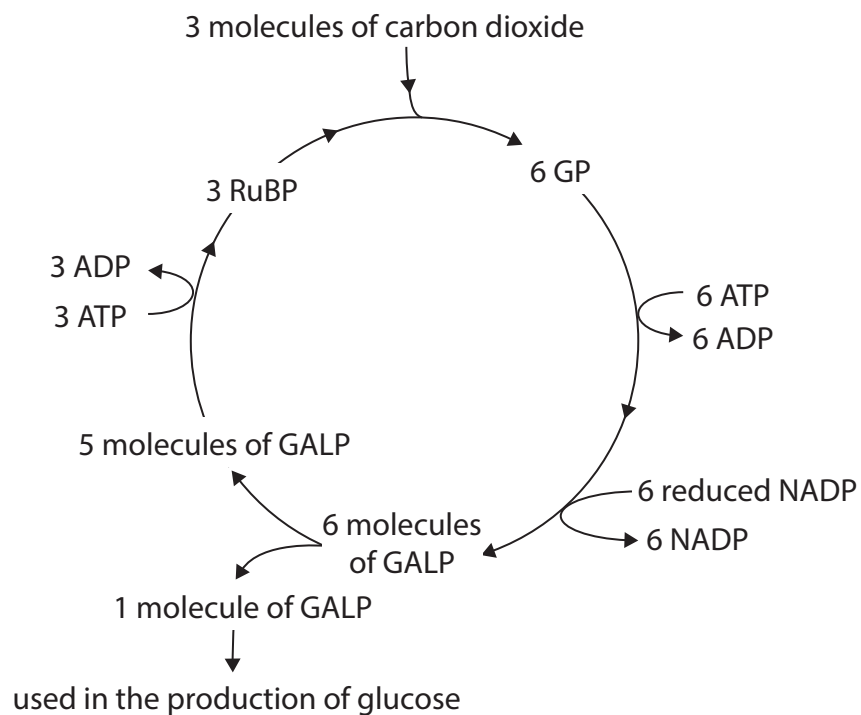
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(b) In the light-dependent reactions, ATP and reduced NADP are generated.

These products are used in the light-independent reactions.

The diagram shows some of the light-independent reactions.



(i) Which describes how ATP is generated in the light-dependent reactions?

(1)

- ☐ **A** by hydrolysis of ADP releasing energy
- ☐ **B** by hydrolysis of ADP requiring energy
- ☐ **C** by phosphorylation of ADP releasing energy
- ☐ **D** by phosphorylation of ADP requiring energy

- (ii) When one molecule of glucose is made in the light-independent reactions, the ratio of ATP used to reduced NADP used is 1.5 : 1.

Show how this ratio can be calculated, using the information in the diagram.

(2)

- (iii) Explain why only a small proportion of the GALP produced is used to make the glucose.

(2)

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(Total for Question 4 = 12 marks)